

BRAC UNIVERSITY

Department of Computer Science and Engineering

CSE 260: Digital Logic Design

Examination: Quiz I

Semester : Summer 2025

Duration: 20min

Full Marks: 15

Name:	ID:	Section:
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1	In a legacy computing system, memory addresses were originally stored in octal. After a recent upgrade, addresses are entered in base 12 (using digits 0–9 with A representing 10 and B representing 11), then converted to decimal and then to hexadecimal to determine the physical location, and finally represented in octal for compatibility with legacy hardware. Given the base-12 memory address $(A3B.82)_{12}$, compute its equivalent octal representation for the given legacy system. Show calculations for each step. [2+2+2+2] Calculating up to 2 fractional points is enough.	6
2	Let: $X = (146)_8, Y = (73)_8, Z = (25)_8, W = (12)_8$ - Evaluate the expression in octal: $E = (X * Y) + (Z - W)$ [3] - Convert E into BCD. [1]	4
3	Perform multiplication of $(25471245)_7$ and $(246)_7$ in base 7. Verify your result by converting to decimal. [2]	2
4	Add -50 and -85 using 8-bit 2's complement. State if there is an overflow or not. [3]	3